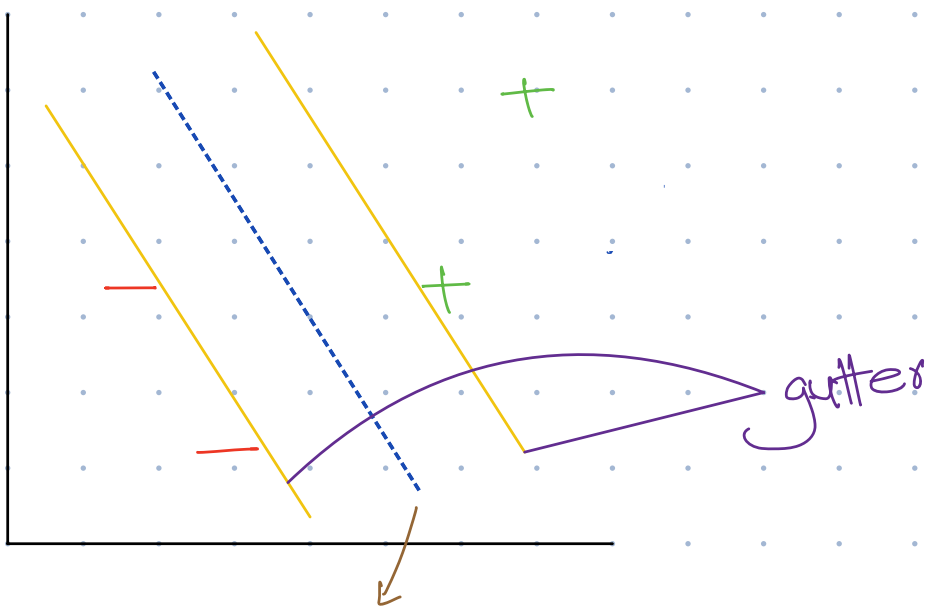
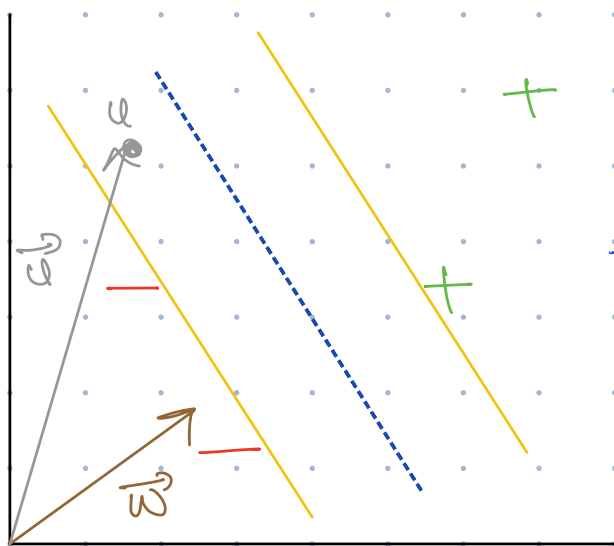




we draw this line with a view toward putting in the widest street that separates the +ve & -ve samples

Now, imagine a vector $\vec{w}$ that is perpendicular to the median of line of the street.

Further, we have an unknown point u & a vector $\vec{u}$ that points to it.



We project $\vec{u}$ onto $\vec{w}$ such that we will distance in $\vec{w}$'s direction & the bigger it is, the closer it will be on right side of the street. Mathematically,

$$\vec{w} \cdot \vec{u} \geq c$$

threshold for being + sample

$$\vec{w} \cdot \vec{u} + b \geq 0 \quad \text{Then '+' decision rule}$$

where $c = -b$

Lets say we have '+' sample

$$\vec{w} \cdot \vec{X}_+ + b \geq 1 \quad (\text{This function enforces that if } x \text{ is}$$

↪ ① a + sample, then the dot product is ≥ 1

Similarly, if we have '-' sample

$$\vec{w} \cdot \vec{x}_- + b \leq -1 \text{ --- ②}$$

For mathematical convenience, we introduce a new variable y_i such that $y_i = +1$ for + samples
 $y_i = -1$ for - samples

Now, write above equations as:

$$y_i (\vec{w} \cdot \vec{x}_i + b) \geq 1 \text{ (from ①)}$$

$$-(\vec{w} \cdot \vec{x}_i + b) \geq 1 \text{ (from mult both sides in ② by -1)}$$

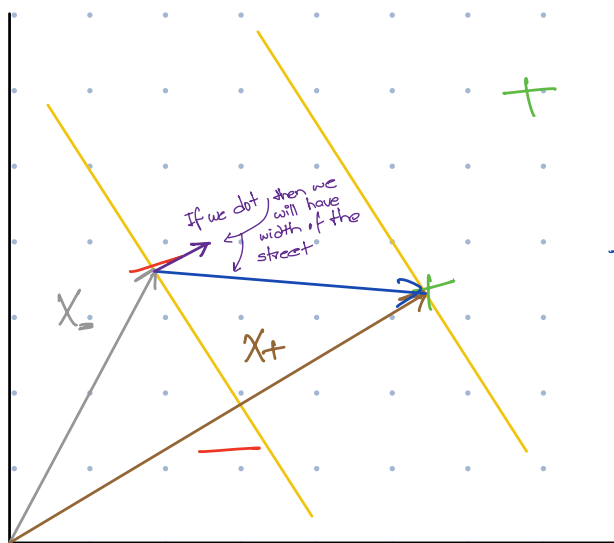
$$y_i (\vec{w} \cdot \vec{x}_i + b) \geq 1$$

Now,

$$y_i (\vec{w} \cdot \vec{x}_i + b) - 1 \geq 0$$

For x_i on the gutter

$$y_i (\vec{w} \cdot \vec{x}_i + b) - 1 = 0 \text{ (3) (constraint)}$$



$$\text{width} = (x_+ - x_-) \cdot \vec{w} \quad \left. \vphantom{\text{width}} \right\} \text{Projects the difference}$$

$$\frac{1}{\|\vec{w}\|}$$

vector onto unit vector
which gives width of
the street.

The difference vector
can point in any direction

$$= \left(X_+ \vec{w} - X_- \vec{w} \right) \frac{1}{\|\vec{w}\|}$$

$$= (1-b + 1+b) \frac{1}{\|\vec{w}\|}$$

From eq (3):

From eq (3):

$$w x_+ + b - 1 = 0$$

$$-w x_- - b - 1 = 0$$

$$w x_+ = 1 - b$$

$$-w x_- = 1 + b$$

$$= \frac{2}{\|\vec{w}\|} \quad \left\{ \begin{array}{l} \text{we want to maximize} \\ \text{this} \end{array} \right.$$

So,

$$\max \frac{1}{\|\vec{w}\|}$$

$$\min \|\vec{w}\|$$

$$\min \frac{1}{2} \|\vec{w}\|^2$$

So, we want to minimize above while honoring the
constraint (3)

In 1802, we learned that if we want to find extremum
of a function with constraint then we need to use
Lagrange's multipliers.

$$L = \frac{1}{2} \|\vec{w}\|^2 - \sum \alpha_i \left[y_i (\vec{w} \cdot \vec{x}_i + b) - 1 \right]$$

$$\frac{\partial L}{\partial \vec{w}} = \vec{w} - \sum \alpha_i y_i \vec{x}_i = 0$$

$$\vec{w} = \sum \alpha_i y_i \vec{x}_i$$

$$\frac{\partial L}{\partial b} = -\sum \alpha_i y_i = 0$$

$$\sum \alpha_i y_i = 0$$

$$L = \frac{1}{2} \left(\sum \alpha_i y_i \vec{x}_i \right) \left(\sum \alpha_j y_j \vec{x}_j \right)$$

→

$$\sum \alpha_i y_i x_i \left(\sum \alpha_j y_j x_j \right) - \underbrace{\sum \alpha_i y_i}_b b + \sum \alpha_i$$

$$= \sum \alpha_i - \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j y_i y_j x_i \cdot x_j$$

Now,

$$\sum \alpha_i y_i \underbrace{x_i \cdot \vec{u}}_{\text{dot product}} + b \geq 0 \text{ Then } +$$

dot product
depends on
sample vector
Δ unknown

SVM never gets stuck in local maximum as this space is a convex.

If samples are linearly separable, then we move them to another space.

So, we want a transformation $\Phi(\vec{x})$

Now, we want to maximize

$$\Phi(x_i) \cdot \Phi(\vec{x}_j)$$

$$\Phi(x_i) \cdot \Phi(u)$$

Put it as a function (kernel trick)

$$k(x_i, x_j) = \Phi(\vec{x}_i) \cdot \Phi(\vec{x}_j)$$

The Dataset

Point (i)	x_1	x_2	Label (y_i)
1	1	2	+1
2	2	1	-1
3	0	0	+1
4	1	0	-1
5	0	1	+1

 Export to Sheets



Step 1: The Dot Product Matrix ($x_i \cdot x_j$)

The first step in an SVM is calculating the similarity between every pair of points using the dot product: $x_i \cdot x_j = (x_{i,1} \cdot x_{j,1}) + (x_{i,2} \cdot x_{j,2})$. ⌂+3

- **Row 1 (x_1):**
 - $x_1 \cdot x_1 = (1 \cdot 1) + (2 \cdot 2) = 1 + 4 = 5$
 - $x_1 \cdot x_2 = (1 \cdot 2) + (2 \cdot 1) = 2 + 2 = 4$
 - $x_1 \cdot x_3 = (1 \cdot 0) + (2 \cdot 0) = 0$
 - $x_1 \cdot x_4 = (1 \cdot 1) + (2 \cdot 0) = 1$
 - $x_1 \cdot x_5 = (1 \cdot 0) + (2 \cdot 1) = 2$
- **Row 2 (x_2):**
 - $x_2 \cdot x_2 = (2 \cdot 2) + (1 \cdot 1) = 5$
 - $x_2 \cdot x_3 = (2 \cdot 0) + (1 \cdot 0) = 0$
 - $x_2 \cdot x_4 = (2 \cdot 1) + (1 \cdot 0) = 2$
 - $x_2 \cdot x_5 = (2 \cdot 0) + (1 \cdot 1) = 1$
- **Row 3 (x_3):**
 - $x_3 \cdot x_3 = (0 \cdot 0) + (0 \cdot 0) = 0$
 - $x_3 \cdot x_4 = (0 \cdot 1) + (0 \cdot 0) = 0$
 - $x_3 \cdot x_5 = (0 \cdot 0) + (0 \cdot 1) = 0$
- **Row 4 (x_4):**
 - $x_4 \cdot x_4 = (1 \cdot 1) + (0 \cdot 0) = 1$
 - $x_4 \cdot x_5 = (1 \cdot 0) + (0 \cdot 1) = 0$
- **Row 5 (x_5):**
 - $x_5 \cdot x_5 = (0 \cdot 0) + (1 \cdot 1) = 1$

$$y_i \left(\left[\sum_{j=1}^n \alpha_j y_j (x_j \cdot x_i) \right] + b \right) = 1$$

Step 2: The Decision Equations

We solve for the Lagrange Multipliers α_i . The rules for a point to be a **Support Vector** (on the "gutter") are: $\mathcal{O}+2$

1. $\sum \alpha_i y_i = 0$ (The "Sum Constraint") $\mathcal{O}+1$
2. $y_i(\sum \alpha_j y_j (x_i \cdot x_j) + b) = 1$ $\mathcal{O}$

Let's pick the 3 points closest to the boundary to be our Support Vectors: P_1, P_2, P_5 . (We assume $\alpha_3 = 0$ and $\alpha_4 = 0$ for the dry run).

Equation A (The Sum Constraint): $\alpha_1(y_1) + \alpha_2(y_2) + \alpha_5(y_5) = 0$

$$\alpha_1(1) + \alpha_2(-1) + \alpha_5(1) = 0 \implies \alpha_2 = \alpha_1 + \alpha_5 \quad \mathcal{O}$$

Equation B (Boundary for P_1): $1 \cdot [\alpha_1(1)(x_1 \cdot x_1) + \alpha_2(-1)(x_2 \cdot x_1) + \alpha_5(1)(x_5 \cdot x_1) + b] = 1$
 $5\alpha_1 - 4\alpha_2 + 2\alpha_5 + b = 1$

Equation C (Boundary for P_2): $-1 \cdot [\alpha_1(1)(x_1 \cdot x_2) + \alpha_2(-1)(x_2 \cdot x_2) + \alpha_5(1)(x_5 \cdot x_2) + b] = 1$
 $-4\alpha_1 + 5\alpha_2 - 1\alpha_5 - b = 1$

Equation D (Boundary for P_5): $1 \cdot [\alpha_1(1)(x_1 \cdot x_5) + \alpha_2(-1)(x_2 \cdot x_5) + \alpha_5(1)(x_5 \cdot x_5) + b] = 1$
 $2\alpha_1 - 1\alpha_2 + 1\alpha_5 + b = 1$

Step 3: Solving the System (Substitution)

First, let's substitute $\alpha_2 = \alpha_1 + \alpha_5$ into B, C, and D:

- **Eq B:** $5\alpha_1 - 4(\alpha_1 + \alpha_5) + 2\alpha_5 + b = 1 \implies \alpha_1 - 2\alpha_5 + b = 1$
- **Eq C:** $-4\alpha_1 + 5(\alpha_1 + \alpha_5) - \alpha_5 - b = 1 \implies \alpha_1 + 4\alpha_5 - b = 1$
- **Eq D:** $2\alpha_1 - (\alpha_1 + \alpha_5) + \alpha_5 + b = 1 \implies \alpha_1 + b = 1$

Now we solve these 3 bolded equations:

1. From Eq D: $b = 1 - \alpha_1$
2. Substitute b into Eq B: $\alpha_1 - 2\alpha_5 + (1 - \alpha_1) = 1 \implies -2\alpha_5 = 0 \implies \alpha_5 = 0$.
3. Substitute $\alpha_5 = 0$ into Eq C: $\alpha_1 + 4(0) - b = 1 \implies \alpha_1 - b = 1$
4. Now we have two simple ones:
 - $\alpha_1 + b = 1$
 - $\alpha_1 - b = 1$ Adding them gives $2\alpha_1 = 2 \implies \alpha_1 = 1$. Subtracting gives $2b = 0 \implies b = 0$.
5. Solve for α_2 : $\alpha_2 = \alpha_1 + \alpha_5 = 1 + 0 = 1$.

Final Multipliers: $\alpha_1 = 1, \alpha_2 = 1, \alpha_5 = 0$ (Note: P_5 ended up not being needed).

Step 4: Calculate Weights (w)

$$w = \sum \alpha_i y_i x_i \quad w = (1)(+1) \begin{bmatrix} 1 \\ 2 \end{bmatrix} + (1)(-1) \begin{bmatrix} 2 \\ 1 \end{bmatrix} + (0)(+1) \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad w = \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad \mathcal{O}+1$$

Step 5: Final Decision Rule

Step 5: Final Decision Rule

$$w_1x_1 + w_2x_2 + b = 0 \implies -1(x_1) + 1(x_2) + 0 = 0 \implies \mathbf{x_2 = x_1}$$

Step 6: Calculate Margin Width

$$\text{Width} = \frac{2}{\|w\|} \quad \|w\| = \sqrt{(-1)^2 + 1^2} = \sqrt{2} \quad \text{Width} = \frac{2}{\sqrt{2}} = \sqrt{2} \approx \mathbf{1.41}$$

This width is the **generalization buffer** we talked about—it's the empty "street" that keeps the model from being confused by noise.